MIL-M-48767(AR)
AMENDMENT 3
1 November 1983
SUPERSEDING
AMENDMENT 2
17 December 1979

MILITARY SPECIFICATION

METHYL FORMATE
(FOR USE IN AMMUNITION)

This Amendment forms a part of Military Specification MIL-M-48767(AR), dated 19 June 1975, and is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and AGencies of the Department of Defense.

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Delete "(PA)" wherever it appears and substitute "(AR)".

Delete the Preamble and substitute the following:

"This specification is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense".

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* 3.1.3 Purity. Delete this table in its entirety and substitute the following:

"Impurity	Maximum Limit, ppm
Water	100
Methanol	100
Dimethyl Ether	50
Formic Acid	50
Halogens:	
Inorganic Halogen	20
Organic Halogen	30
Total Impurities	150"

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4.4.2 Testing. To end of sentence, add the following: "at a Relative Humidity not to exceed 4% at ambient conditions."

FSC 6810

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4.4.2.1 Sampling. At the end of the second sentence add the following: "or the sample from each batch that is combined into a shipping lot shall be subjected to the tests listed below in accordance with MIL-STD-105, Level II."

Delete Inspection/Requirements table and substitute the following table:

"Inspection/Requirement	Defect Classification
Form (see 3.1.1)	Major
Infrared Spectroscopy (see 3.1.2)	Major
Purity (see 3.1.3)	Major"
Inorganic Halogens	
Organic Halogens	
Water	
Methanol	
Dimethyl Ether	
Formic Acid	
Total Impurities	

4.4.3.3.2 After "Other Impurities" add the following: "(Not Including Halogens)"

* 4.4.3.3.3 Halogen. Delete in its entirety and substitute the following:

"4.4.3.3.3 Halogen determination.

4.4.3.3.3.1 Inorganic halogen. The procedure outlined below is the Turbidimetric analytical technique. The colorimetric or nephelometric standard analytical techniques may also be used.

a. Equipment for Turbidimetric method.

(1) Turbidimeter (Hach Chemical Model 2100A or equivalent).

(2) Reagents: Sodium Chloride solution (5 μg/ml)
Silver nitrate solution (5%)
Nitric Acid (concentrated)

b. Procedure for turbidimetric method:

(1) Make up standard halogen solutions as follows:

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Place 0, 1, 2, and 3 ml of 5 μg Cl /ml sodium chloride solution respectively into four 50 ml flasks, each of which contains 1 ml of concentrated HNO_3 and 1 ml of 5% $AgNO_3$. Fill up to the mark with water, shake well, let them age in a dark place for 10 minutes, and measure turbidity. Plot a standard curve of halogen concentration against turbidity.

(2) In a beaker, dissolve 0.5 ml of methyl formate in 15 ml of water.

(3) In a separate container, make up a solution of diluted nitric acid using 27 ml nitric acid and 9 ml water.

(4) Transfer the methyl formate solution to a 50 ml volumetric flask and rinse the beaker with three 11 ml portions of the nitric acid solution. Add the rinsings to the volumetric flask. The total volume of solution will now be approximately 48 ml.

(5) Measure turbidity, and use this as the blank value. The turbidity of this solution is normally as low as that of deionized water.

(6) Add 1 ml of 5% $AgNO_3$ to the flask, and fill to the 50 ml mark with water.

(7) Shake well and let stand in a dark place for 10 minutes.

(8) Measure turbidity and determine the total halogen content by using the standard curve.

4.4.3.3.3.2 Organic halogen.

a. Equipment:

(1) Gas chromatograph (Hewlett-Packard 5840A or equivalent).

(2) Detector - Flame Ionization (FID) or Electron Capture (ECD).

(3) Column - Silicone Gum Capillary (Hewlett-Packard SE54 or equivalent).

NOTE: Stainless steel column, OV-101 (6 meters length, 1/8 inch diameter) may also be used.

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b. Gas chromatograph conditions:

(1) Detector temperature: 290°C

(2) Column temperature: 40°C

(3) Carrier gas: Helium at 60 ml/min.

(4) Injection temperature: 290°C.

(5) Sample size: 1 µl.

c. Procedure:

(1) Inject 1 µl of methyl formate

(2) Compare the peak area with known standards for quantitative analysis, and identify any unknown peaks."

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* 6.4 Add the following source of supply:

"Tedia Company
2940 Symmes Road
Box 342
Fairfield, Ohio 54014"

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

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